Employee Burnout Prediction Using Machine Learning

Project Live Link

GitHub Repository: https://github.com/kano8689/Employee-Burnout

Live Application: https://employeeburnoutml.vercel.app/

1. Abstract

Employee burnout has become one of the most critical challenges faced by organizations across industries. Increasing workloads, tight deadlines, remote working environments, and poor work-life balance contribute significantly to physical and mental exhaustion among employees. Burnout negatively impacts employee productivity, organizational performance, job satisfaction, and retention rates. Traditional methods of identifying burnout, such as employee surveys, manager observations, and periodic performance reviews, often rely on subjective judgment and fail to detect burnout at an early stage. Therefore, organizations require intelligent, data-driven solutions capable of identifying burnout risks accurately and efficiently.

This project presents an Artificial Intelligence and Machine Learning-based Employee Burnout Prediction System that predicts an employee's burnout score using workplace and behavioral attributes. The proposed solution employs the Random Forest Regression algorithm to estimate a continuous burnout score ranging from 0 to 1. Based on the predicted score, employees are automatically categorized into Low, Medium, or High burnout risk levels. Along with prediction, the application also provides recommendations and feature importance analysis, enabling users to understand which workplace factors contribute most to burnout.

The system follows a complete Machine Learning pipeline consisting of data collection, preprocessing, feature engineering, model training, evaluation, deployment, and real-time prediction. Data preprocessing techniques such as missing value handling, categorical feature encoding, feature scaling, and feature selection ensure that the model receives clean and structured input data. The trained Random Forest model demonstrates strong predictive capability while maintaining robustness against noisy data and overfitting. Performance is evaluated using standard regression metrics including R^2 Score, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

To make the model practical and accessible, it is deployed as a web application using the Flask framework. The application supports both single-employee prediction through an interactive form and bulk prediction through CSV file uploads. Users receive real-time predictions accompanied by an animated speedometer gauge, burnout classification, feature contribution percentages, and personalized recommendations. Additionally, a dedicated Model Insights page displays evaluation metrics, feature importance rankings, and dataset information to improve transparency and confidence in the model.

Overall, this project demonstrates how Machine Learning can be effectively integrated with modern web technologies to support Human Resource Analytics. The developed application provides an intelligent decision-support tool that enables organizations to identify employees at risk of burnout proactively, allowing timely intervention and promoting a healthier, more productive workplace. The system also serves as a scalable foundation for future enhancements such as Explainable AI (XAI), cloud integration, and real-time HR database connectivity. The report builds on the existing project documentation while incorporating the latest application features and architecture.

2. Introduction

Employee well-being has emerged as a key determinant of organizational success in today's competitive business environment. While organizations increasingly emphasize productivity and performance, employees often face high workloads, continuous deadlines, extended working hours, and growing workplace pressure. Over time, these conditions can lead to employee burnout—a state of emotional, physical, and mental exhaustion that reduces motivation, job satisfaction, and overall performance. If left unaddressed, burnout may result in decreased productivity, increased absenteeism, higher employee turnover, and significant financial losses for organizations.

Traditionally, organizations rely on employee feedback surveys, managerial observations, counseling sessions, or periodic performance evaluations to assess burnout. Although these methods provide valuable insights, they are time-consuming, subjective, and often incapable of detecting burnout before it significantly affects employee performance. Human judgment may vary across managers, and manual evaluation becomes increasingly difficult as organizations grow larger. Consequently, many employees remain unnoticed until burnout has already impacted their productivity and well-being.

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have transformed the way organizations analyze employee-related data. Instead of relying solely on human observations, ML algorithms can identify hidden relationships between multiple workplace factors and employee burnout by learning from historical data. These predictive models provide objective, data-driven insights that enable organizations to identify at-risk employees before severe burnout occurs. Early detection allows Human Resource (HR) departments to implement appropriate interventions such as workload balancing, flexible work arrangements, wellness programs, or counseling support.

The Employee Burnout Prediction Using Machine Learning project aims to address these challenges by developing a smart web-based prediction system capable of estimating employee burnout accurately. The system utilizes workplace attributes such as gender, company type, work-from-home availability, designation level, resource allocation, and mental fatigue score to predict a continuous burnout score between 0 and 1. Unlike traditional classification systems that only identify risk categories, this approach provides both an exact burnout score and meaningful burnout classifications (Low, Medium, or High), offering greater flexibility and interpretability.

The core prediction engine is built using the Random Forest Regression algorithm, an ensemble Machine Learning technique known for its high prediction accuracy, robustness against noisy data, and ability to estimate feature importance. One of the major advantages of Random Forest is its capability to determine how strongly each input feature contributes to the final prediction. This functionality enables the application to present users with a Feature Impact (%) visualization, making the prediction process more transparent and understandable.

To ensure practical usability, the Machine Learning model is integrated into a Flask-based web application featuring an intuitive and responsive user interface. The application supports both individual employee prediction through an interactive web form and bulk prediction through CSV file uploads, allowing HR professionals to evaluate large employee datasets efficiently. Additional features such as an animated burnout speedometer, personalized recommendations, model evaluation metrics, and a Model Insights dashboard further enhance the user experience.

The project follows a complete end-to-end Machine Learning workflow, beginning with data preprocessing and feature engineering, followed by model training, evaluation, deployment, and real-time prediction. It demonstrates how modern AI technologies can be applied to solve real-world Human Resource Analytics problems while maintaining scalability, usability, and prediction transparency. Compared to conventional burnout assessment methods, the proposed system significantly reduces manual effort, improves prediction consistency, and provides actionable insights for organizational decision-making.

Furthermore, the project has been successfully deployed online using Flask and Vercel, making it accessible through any modern web browser without requiring local installation. This deployment highlights the practical applicability of Machine Learning models beyond experimental environments and demonstrates how AI-powered decision-support systems can assist organizations in maintaining healthier, more productive workforces.

3. Methodology

The Employee Burnout Prediction System follows a structured Machine Learning pipeline to transform raw employee data into meaningful burnout predictions. The overall workflow consists of data collection, preprocessing, model training, evaluation, and deployment.

3.1 Workflow

Dataset
↓
Data Cleaning
↓
Label Encoding
↓
Feature Scaling
↓
Feature Engineering
↓
Train-Test Split
↓
Random Forest Regression
↓
Model Evaluation
↓
Prediction

3.2 Dataset

The project uses an Employee Burnout dataset containing employee demographic and workplace-related information. The dataset includes six input features and one target variable (Burn Rate).

Feature	Type	Description
Gender	Categorical	Male/Female
Company Type	Categorical	Product/Service
WFH Setup	Categorical	Yes/No

Feature	Type	Description
Designation	Numeric	Job level
Resource Allocation	Numeric	Workload
Mental Fatigue	Numeric	Fatigue score

3.3 Data Preprocessing

Step	Purpose
Missing Value Handling	Remove incomplete records
Label Encoding	Convert text to numeric
StandardScaler	Normalize numerical values
Feature Selection	Keep important attributes

3.4 Model Training Process

1. Split dataset (80:20)
2. Train Random Forest
3. Evaluate model
4. Save model
5. Deploy

4. Implementation & Results

The Employee Burnout Prediction System was successfully implemented as a web-based application using Python, Flask, HTML, CSS, and JavaScript. The system integrates a trained Random Forest Regression model with an interactive user interface to provide real-time burnout predictions. It supports both single employee prediction and bulk CSV analysis while presenting prediction results through visual indicators, feature importance, and personalized recommendations. The following sections describe the major application modules and demonstrate the overall functionality and performance of the developed system.

4.1 Single Prediction

Description

The Single Prediction module allows users to enter employee details manually. After preprocessing, the Random Forest model predicts the burnout score, displays the burnout level, feature importance, and recommendations.

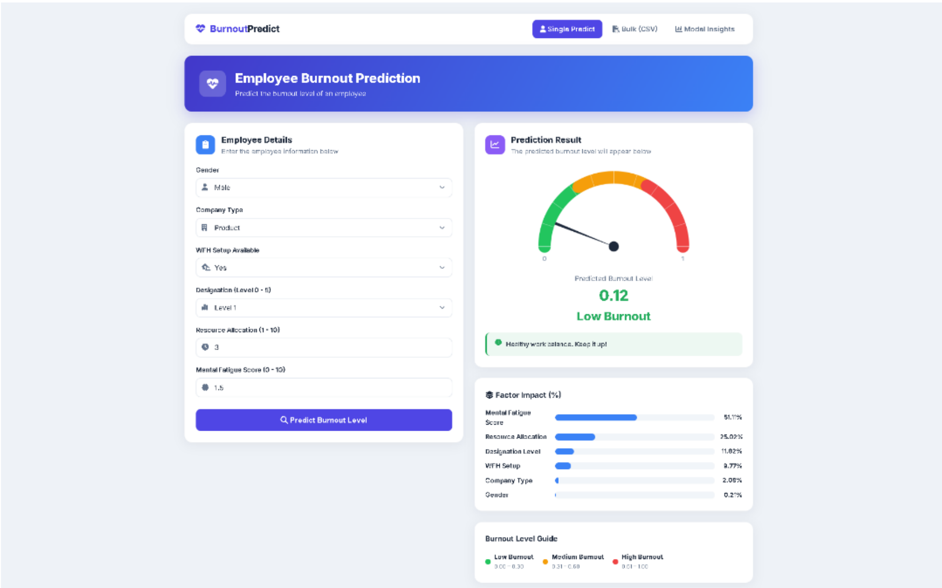

Figure 3: Single record prediction.

4.2 Bulk Prediction

Description

Users can upload a CSV file containing multiple employee records. The system predicts burnout scores for every employee and generates a summary dashboard.

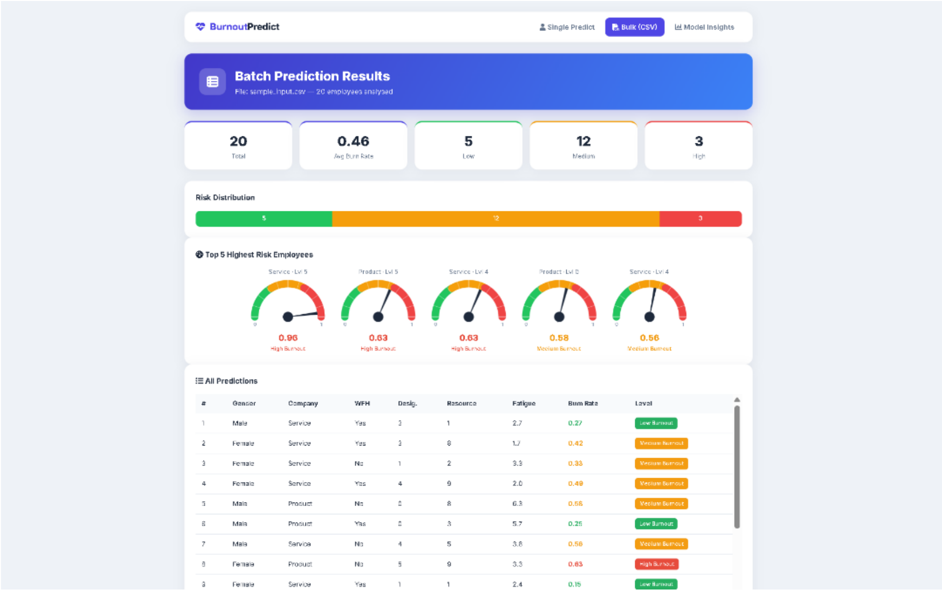

Figure 4: Bulk records from .CSV file prediction.

4.3 Model Insights

Description

The Model Insights page displays R^2 Score, RMSE, MAE, dataset information, and feature importance to evaluate model performance.

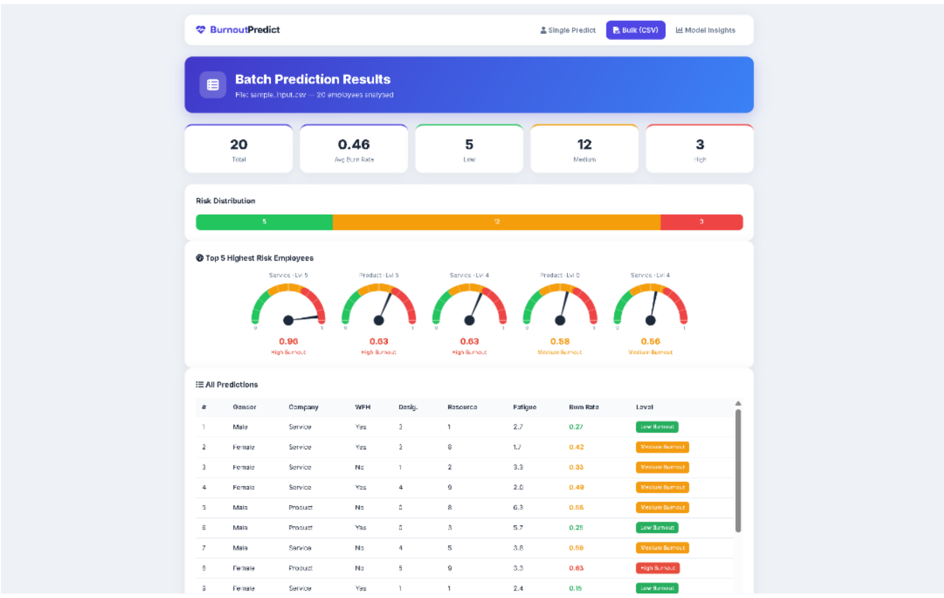

Figure 5: Bulk records from .CSV file prediction

5. Conclusion

The Employee Burnout Prediction System successfully demonstrates how Machine Learning can be applied to support Human Resource Analytics by identifying employees who may be at risk of burnout. Using a Random Forest Regression model, the system analyzes workplace-related factors to predict a burnout score and classify employees into Low, Medium, or High risk levels. The application provides a practical, data-driven approach for assisting organizations in monitoring employee well-being and making informed decisions.

The project integrates the complete Machine Learning workflow, including data preprocessing, feature engineering, model training, evaluation, and deployment into a Flask-based web application. Features such as single employee prediction, bulk CSV prediction, feature importance analysis, an animated speedometer gauge, personalized recommendations, and a Model Insights dashboard enhance the usability and transparency of the system. The deployed web application delivers real-time predictions through an intuitive and user-friendly interface.

Overall, this project demonstrates the successful combination of Artificial Intelligence, Machine Learning, and web development to solve a real-world workplace problem. With future enhancements such as HRMS integration, Explainable AI (SHAP/LIME), cloud deployment, and continuous model retraining, the system can be further improved into a comprehensive employee wellness and decision-support platform.

6. References

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